Explainable Deep Learning for Deepfake Voice Detection: A Leakage-Free Re-evaluation and Cross-Dataset Study

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***Abstract*—We revisit a recently reported machine-learning approach to deepfake voice detection that claims 99.93% accuracy, and show that this figure stems from a leaky, small-scale evaluation rather than genuine detection ability. We merge three public corpora—ASVspoof 2021, a LibriSpeech/TTS set, and MLAAD v5—into a single balanced benchmark, and re-evaluate under a speaker-disjoint protocol using Equal Error Rate (EER). On this honest benchmark we reproduce the paper's classical pipeline as a baseline (best 11.33% EER) and propose *AttentiveSpecCNN*, a compact convolutional network with a learnable temporal-attention pooling layer that both improves accuracy and yields built-in explanations via attention weights and Grad-CAM. The proposed model reaches 4.81% EER, a ∼58% relative reduction over the baseline. Finally, a leave-one-source-out study shows that this in-domain success does *not* transfer to a fully unseen source, quantifying an important and often-overlooked generalisation gap.**

***Index Terms*—audio deepfake detection, anti-spoofing, explainable AI, attention, EER, cross-dataset generalisation.**

I. Introduction

Synthetic speech is now realistic enough to threaten voice authentication and enable fraud and misinformation, motivating reliable detectors. A recent study, "Unmasking the Fake" [1], reports 99.93% accuracy using Mel Frequency Cepstral Coefficients (MFCC) refined by a Gaussian-Naive-Bayes and Non-negative-Matrix-Factorisation "transfer" step feeding classical classifiers.

Such a figure is not comparable to properly evaluated anti-spoofing systems. It is obtained on a small, single Kaggle set using random k-fold cross-validation and synthetic minority oversampling applied *before* splitting, both of which leak information between train and test; the reported per-class metrics are internally inconsistent; and no EER—the field-standard metric—is given. We therefore treat the method as a baseline to reproduce rather than a target.

Our contributions are: (i) a unified, cleaned, class-balanced benchmark merged from three public corpora; (ii) a rigorous *speaker-disjoint* evaluation with EER; (iii) *AttentiveSpecCNN*, an explainable model that replaces the paper's static feature engineering with a learnable attention mechanism and improves EER by ∼58% relative over the reproduced baseline; and (iv) a cross-dataset study exposing the limits of in-domain performance.

TABLE I
The three merged source datasets. The combined corpus is ∼97% fake; a class-balanced subset (19,251 / 19,251) is used.

Source (local name)	Description	Real	Fake
ASVspoof 2021 (DF)	Telephone/codec deepfakes	16,977	441,894
Adarsh AD set	LibriSpeech real + TTS fakes	2,274	2,173
MLAAD v5	Multilingual TTS (fake only)	0	151,446
Merged	Unified manifest	**19,251**	**595,513**

II. Datasets and Preprocessing

We merge three public sources (Table I): the ASVspoof 2021 deepfake (DF) track [2], a set pairing LibriSpeech real speech with modern text-to-speech (TTS) fakes, and MLAAD v5 [3], a multilingual TTS corpus spanning roughly ninety generators. The combined corpus is ∼97% fake, so we draw a class-balanced subset (19,251 real / 19,251 fake) whose fake half is spread evenly across all sources and generators to avoid a single-generator bias. Audio is decoded, converted to mono, resampled to 16 kHz, fixed to 4 s, and transformed to 80-band log-mel spectrograms.

III. Method

A. *AttentiveSpecCNN*

The proposed detector is a compact (≈106K-parameter) 2-D CNN over log-mel spectrograms followed by a learnable temporal-attention pooling layer. Four convolutional blocks (channels 16/32/64/128, each 3×3 conv, batch norm, ReLU and 2×2 max-pool) produce a time–frequency feature map; the frequency axis is averaged and an attention layer computes a softmax weight per time frame, forming a context vector classified by a small multilayer perceptron. This attention layer is the key departure from [1]: rather than a fixed MFCC→GNB→NMF transform, it *learns* how much each moment of audio contributes to the decision. Table II lists the configuration.

TABLE II
AUDIO FRONT-END, MODEL, AND TRAINING CONFIGURATION.

Setting	Value
Sample rate / channels	16 kHz, mono
Clip length	4 s (64,000 samples); pad, else random (train) / center (eval) crop
Log-mel	80 bands, n_{fft}=400, hop=160 $\rightarrow$ 80×401
CNN backbone	4 blocks, 16/32/64/128, 3×3 conv+BN+ReLU+2×2 pool
Pooling	frequency mean, then temporal attention
Head	Linear 128→64, ReLU, dropout 0.3, Linear 64→2
Parameters	≈106,000
Optimiser	Adam, lr 10^{-3}, ReduceLROnPlateau (×0.5, patience 2)
Loss	cross-entropy, inverse-frequency class weights
Batch / epochs	64 / 15, best-val-EER checkpoint
Hardware	single NVIDIA RTX 3080

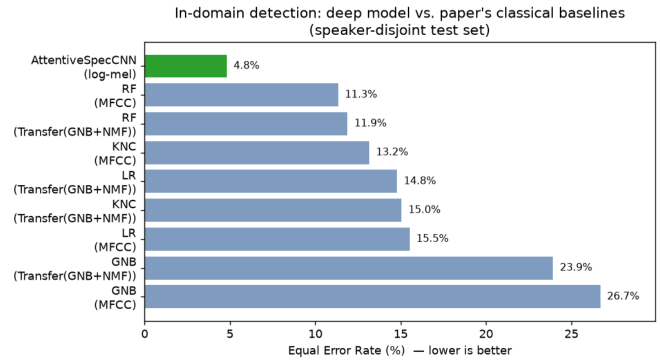

Fig. 1. Equal Error Rate by method; the proposed model roughly halves the best classical baseline's EER.

B. Explainability

Two saliency views come for free. The attention weights give a per-frame importance curve, and Grad-CAM over the final convolutional layer highlights the time–frequency regions that drove the predicted class (Fig. 2).

C. Classical baseline

We reproduce [1] with 40 MFCCs summarised by mean and standard deviation (an 80-D vector), standardised before classification. Random Forest, Logistic Regression, k-Nearest Neighbours and Gaussian Naive Bayes are evaluated on the raw MFCCs and on the paper's GNB+NMF "transfer" features (GNB class probabilities concatenated with a 16-component NMF).

IV. EVALUATION PROTOCOL

Two choices make the numbers trustworthy. First, splits are *speaker-disjoint*: train, validation and test share no speaker or TTS source-reader, so a model cannot inflate its score by memorising voice identity, unlike random k-fold. Second, we report EER—the operating point where false accept and false reject rates are equal—alongside balanced accuracy and per-class F1, so the "predict-everything-fake" failure mode cannot hide behind raw accuracy.

V. RESULTS

A. In-domain detection

Table III and Fig. 1 report all methods on the same speaker-disjoint test set. The proposed model roughly halves the best classical baseline's EER. Notably, the paper's GNB+NMF "transfer" step does not help—it slightly worsens the strongest classical model—and the paper's headline GNB is the weakest configuration.

B. Cross-dataset generalisation

To test whether the model detects transferable artifacts or dataset-specific cues, we train on two sources and test on the held-out third (Table IV, Fig. 3). Performance collapses to, or below, chance. Trained on ASVspoof (telephone/codec) and MLAAD (clean TTS), the model appears to associate clean audio with "fake" and mislabels the clean LibriSpeech real speech of dataset 2 (AUC 0.42, below chance). The strong in-domain result thus leans heavily on recording/domain signatures.

VI. DISCUSSION AND LIMITATIONS

The study supports a three-tier conclusion, each with protocol-correct numbers: the paper's 99.93% accuracy is not reproducible or comparable; on a fair benchmark the proposed explainable model achieves 4.81% EER, a ∼58% relative improvement over the reproduced baseline; and cross-dataset generalisation remains open, so in-domain metrics overstate real-world robustness. Limitations: MLAAD contributes only fake audio, so part of the separation reflects domain differences; the ASVspoof-holdout fold is real-starved (only ∼2.3k real clips to train on); and the cross-dataset numbers are single-run, fixed-threshold probes, for which ROC-AUC is the most robust signal.

VII. CONCLUSION

A leakage-free re-evaluation dispels an implausibly high published accuracy and shows that a compact, explainable attention CNN substantially improves honest in-domain detection. The cross-dataset collapse, however, underscores that robust generalisation—not headline accuracy—is the open problem in audio deepfake detection.

REFERENCES

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TABLE III
IN-DOMAIN DETECTION ON THE SPEAKER-DISJOINT TEST SET (LOWER EER IS BETTER).

Method	Features	EER (%)	Bal. Acc (%)	$\mathrm{F1}_{\mathrm{fake}}$ (%)	AUC
AttentiveSpecCNN (ours)	log-mel	**4.81**	**93.84**	**93.49**	**0.99**
Random Forest	MFCC	11.33	87.51	87.86	0.96
Random Forest	GNB+NMF transfer	11.89	87.60	87.83	0.95
k-Nearest Neighbours	MFCC	13.15	85.46	86.13	0.92
Logistic Regression	GNB+NMF transfer	14.78	84.73	84.09	0.92
k-Nearest Neighbours	GNB+NMF transfer	15.05	83.93	84.67	0.91
Logistic Regression	MFCC	15.52	84.33	84.17	0.91
Gaussian Naive Bayes	GNB+NMF transfer	23.93	76.72	75.77	0.87
Gaussian Naive Bayes (paper headline)	MFCC	26.69	73.05	71.75	0.81

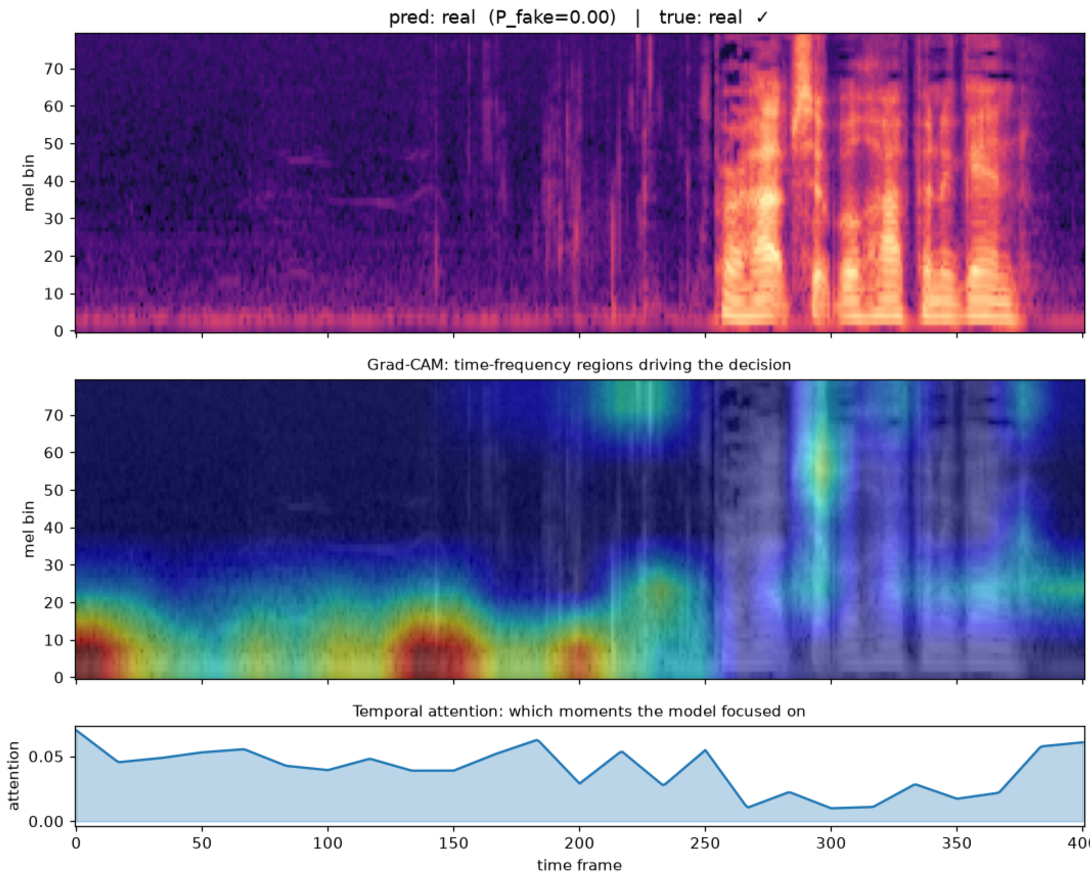

Fig. 2. Explanation for a correctly-identified real clip: log-mel spectrogram (top), Grad-CAM (middle), and temporal attention (bottom).

TABLE IV
LEAVE-ONE-SOURCE-OUT GENERALISATION. MLAAD HAS NO REAL AUDIO, SO ONLY THE FAKE-DETECTION RATE ON UNSEEN GENERATORS IS REPORTABLE.

Held-out source	Trained on	Metric	Result
(ref.) In-domain	all three	EER / AUC	**4.81% / 0.99**
ASVspoof 2021	ds2 + MLAAD	EER / AUC	44.9% / 0.56
dataset 2	ASV + MLAAD	EER / AUC	57.5% / 0.42
MLAAD	ASV + ds2	fake-recall	48.4%

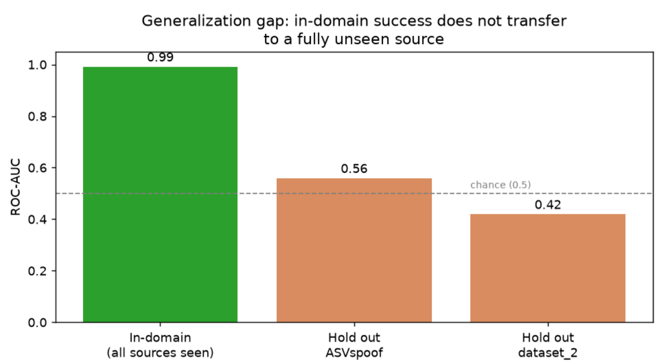

Fig. 3. In-domain vs. held-out ROC-AUC. Performance collapses to (or below) chance on a fully unseen source.